Processing Hive Data

Abinet Joan

2023-12-11 10:29:26 +0100

Contents

Loading packages 1

Loading data 1

QC 2

Normalisation + scalin 3

PCA 3

Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(stringr))
suppressMessages(library(SingleR))
suppressMessages(library(scuttle))
suppressMessages(library(scRNAseq))
suppressMessages(library(RColorBrewer))
suppressMessages(library(formatR))
```

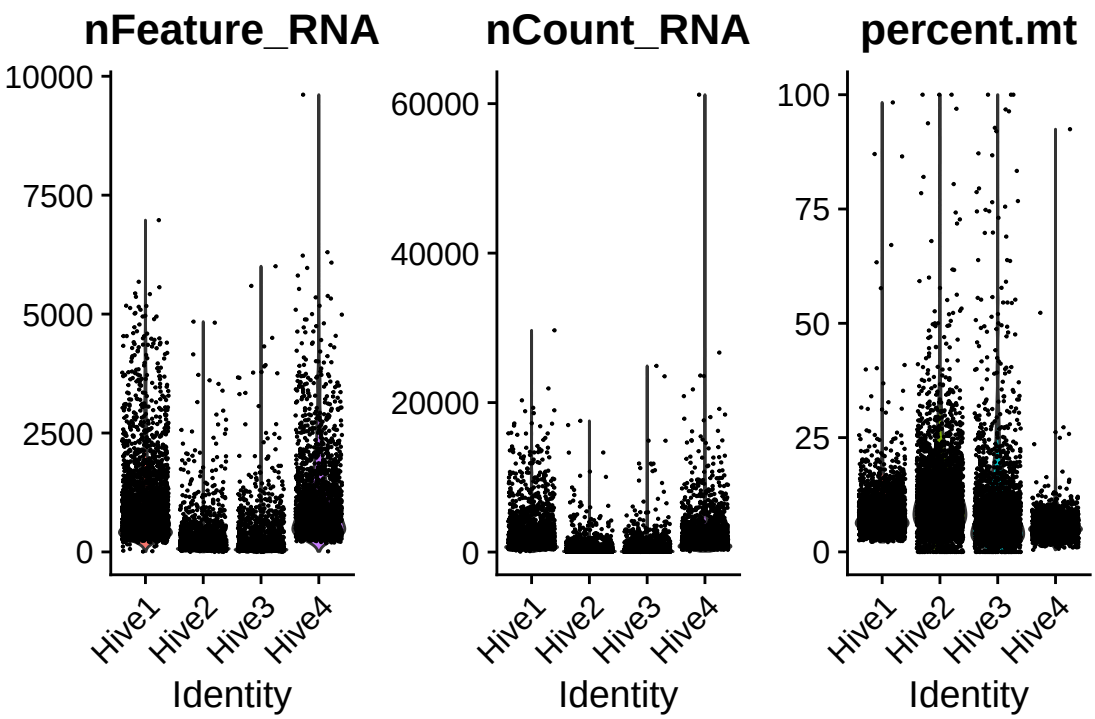
Loading data

```
# The path should be modifie to include your directory of the Hive count matrix
all_dirs <- dir(path = "./Counts", full.names = T)

list_sample <- list()
for (dir in all_dirs) {

  files <- list.files(dir)
  Count_file <- files[grep("TCM.tsv.gz$", files)]
  Countdata <- read.table(paste0(dir,"/",Count_file), sep = "\t", header = T,
row.names=1)

  Lavage_cellsHive <- CreateSeuratObject(counts = Countdata, project =
str_sub(dir,-5,-1))
```

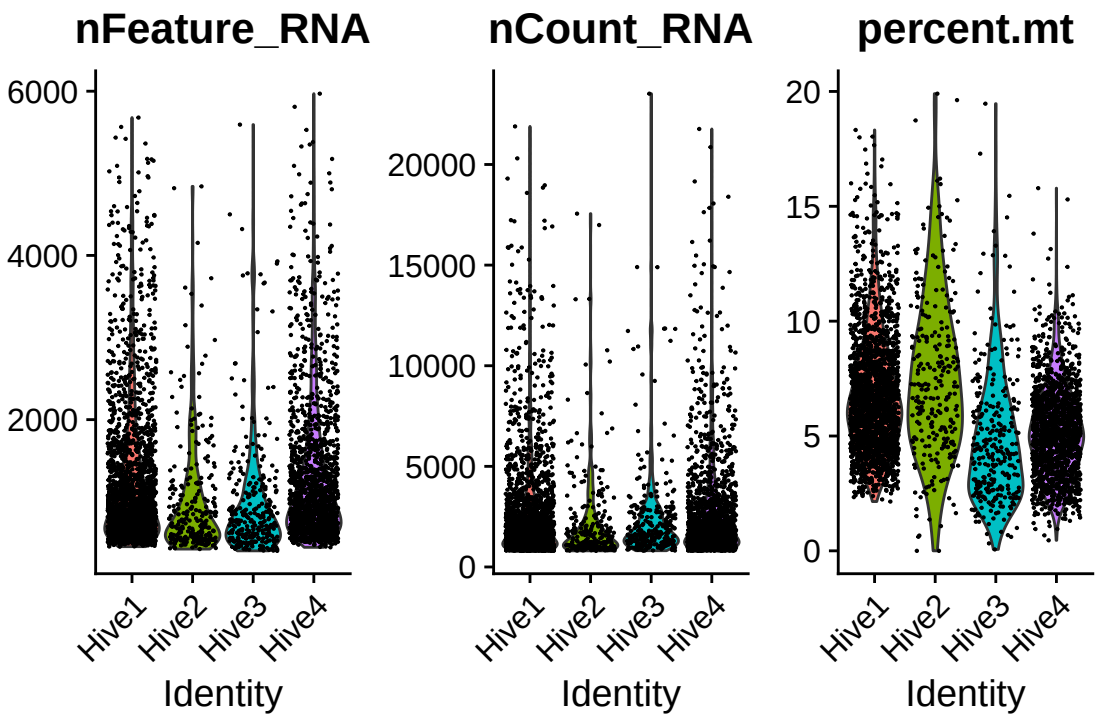
```
  list_sample <- append(list_sample, Lavage_cellsHive)

}

Lavage_cellsHive <- merge(list_sample[[1]], y = list_sample[-1], add.cell.ids =
c("1","2","3", "4"), project = "bronchoalveolar_cells")
```

QC

```
Lavage_cellsHive[["percent.mt"]] <- PercentageFeatureSet(Lavage_cellsHive, pattern =
"^MT-")# MT : human cells
VlnPlot(Lavage_cellsHive, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol
= 3, pt.size = 0.1)
```




```
Lavage_cellsHive <- subset(Lavage_cellsHive, subset = nFeature_RNA < 6000 & nCount_RNA >
800 & percent.mt < 20 & nFeature_RNA > 400 )
VlnPlot(Lavage_cellsHive, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol
= 3, pt.size = 0.1, group.by = "orig.ident")
```



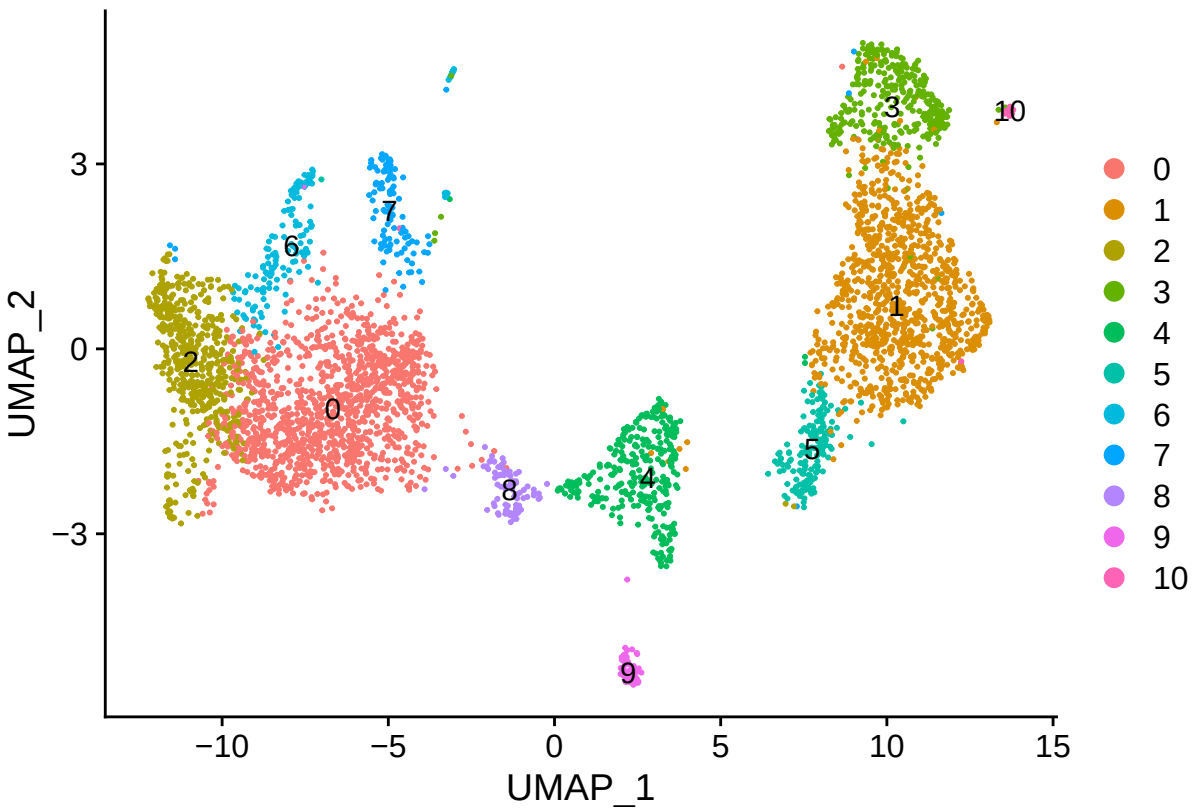

```
#ggsave("plot/QC.png", width = 12, height = 6)
```

Normalisation + scalin

```
Lavage_cellsHive <- NormalizeData(Lavage_cellsHive, normalization.method =
"LogNormalize", scale.factor = 10000)
Lavage_cellsHive <- FindVariableFeatures(Lavage_cellsHive, selection.method = "vst",
nfeatures = 2000)

all.genes <- rownames(Lavage_cellsHive)
Lavage_cellsHive <- ScaleData(Lavage_cellsHive, features = all.genes)
```

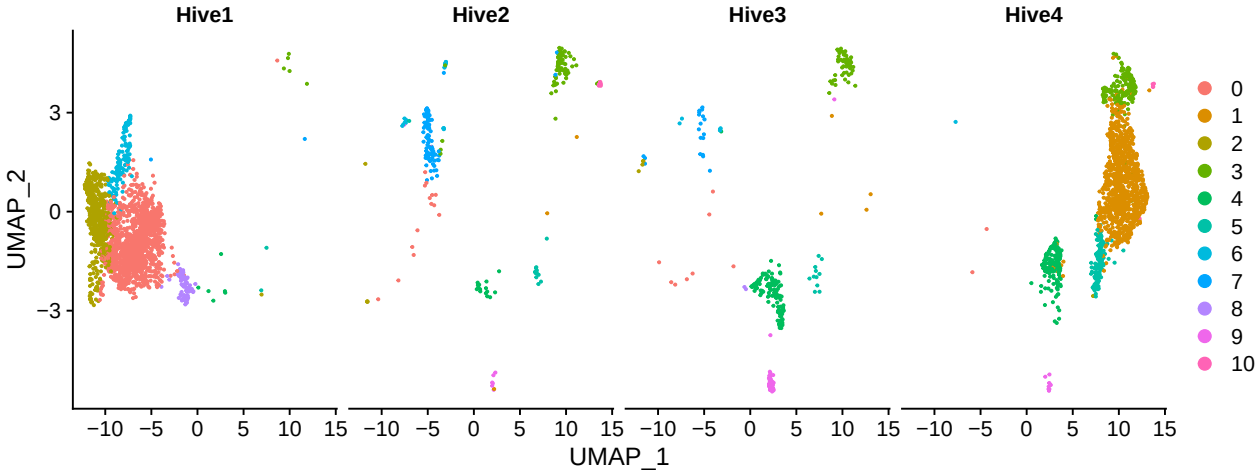
PCA

```
Lavage_cellsHive <- RunPCA(Lavage_cellsHive, features = VariableFeatures(object =
Lavage_cellsHive))

Lavage_cellsHive <- FindNeighbors(Lavage_cellsHive, dims = 1:15)
Lavage_cellsHive <- FindClusters(Lavage_cellsHive, resolution = 0.5)
Lavage_cellsHive <- RunUMAP(Lavage_cellsHive, dims = 1:15)

DimPlot(Lavage_cellsHive, reduction = "umap", label = T)
```




```
#ggsave("plot/umapHive_total.png", width = 12, height = 8)
```

```
DimPlot(Lavage_cellsHive, reduction = "umap", split.by = "orig.ident")
```




```
#ggsave("plot/umapHive.png", width = 18, height = 6)
```

```
saveRDS(Lavage_cellsHive, "Lavage_cellsHive.rds")
```

```
sessionInfo()
```

```
## R version 4.3.2 (2023-10-31)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.3 LTS
```

```
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats4    stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] formatR_1.14                RColorBrewer_1.1-3
##  [3] scRNAseq_2.14.0             scuttle_1.10.1
##  [5] SingleCellExperiment_1.22.0 SingleR_2.2.0
##  [7] SummarizedExperiment_1.30.2 Biobase_2.60.0
##  [9] GenomicRanges_1.52.0        GenomeInfoDb_1.36.0
## [11] IRanges_2.34.0              S4Vectors_0.38.1
## [13] BiocGenerics_0.46.0         MatrixGenerics_1.12.2
## [15] matrixStats_1.0.0           stringr_1.5.0
## [17] ggplot2_3.4.2               patchwork_1.1.2
## [19] SeuratObject_4.1.3          Seurat_4.3.0
## [21] dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] RcppAnnoy_0.0.21              splines_4.3.2
##   [3] later_1.3.1                   BiocIO_1.10.0
##   [5] filelock_1.0.2                bitops_1.0-7
##   [7] tibble_3.2.1                  polyclip_1.10-4
##   [9] XML_3.99-0.14                 lifecycle_1.0.3
##  [11] ensembldb_2.24.0              globals_0.16.2
##  [13] lattice_0.22-5                MASS_7.3-60
##  [15] magrittr_2.0.3                plotly_4.10.2
##  [17] rmarkdown_2.23                yaml_2.3.7
##  [19] httpuv_1.6.11                 sctransform_0.3.5
##  [21] spam_2.9-1                    sp_2.0-0
##  [23] spatstat.sparse_3.0-2         reticulate_1.30
##  [25] cowplot_1.1.1                 pbapply_1.7-2
##  [27] DBI_1.1.3                     abind_1.4-5
##  [29] zlibbioc_1.46.0               Rtsne_0.16
##  [31] purrr_1.0.1                   AnnotationFilter_1.24.0
##  [33] RCurl_1.98-1.12               rappdirs_0.3.3
##  [35] GenomeInfoDbData_1.2.10       ggrepel_0.9.3
##  [37] irlba_2.3.5.1                 listenv_0.9.0
##  [39] spatstat.utils_3.0-3          goftest_1.2-3
```

```
##  [41] spatstat.random_3.1-5         fitdistrplus_1.1-11
##  [43] parallelly_1.36.0             DelayedMatrixStats_1.22.1
##  [45] leiden_0.4.3                  codetools_0.2-19
##  [47] DelayedArray_0.26.3           xml2_1.3.4
##  [49] tidyselect_1.2.0              farver_2.1.1
##  [51] ScaledMatrix_1.8.1            BiocFileCache_2.8.0
##  [53] spatstat.explore_3.2-1        GenomicAlignments_1.36.0
##  [55] jsonlite_1.8.7                ellipsis_0.3.2
##  [57] progressr_0.13.0              ggridges_0.5.4
##  [59] survival_3.5-7                progress_1.2.2
##  [61] tools_4.3.2                   ica_1.0-3
##  [63] Rcpp_1.0.11                   glue_1.6.2
##  [65] gridExtra_2.3                 xfun_0.39
##  [67] withr_2.5.0                   BiocManager_1.30.21
##  [69] fastmap_1.1.1                 fansi_1.0.4
##  [71] digest_0.6.33                 rsvd_1.0.5
##  [73] R6_2.5.1                      mime_0.12
##  [75] colorspace_2.1-0              scattermore_1.2
##  [77] tensor_1.5                    biomaRt_2.56.1
##  [79] spatstat.data_3.0-1           RSQLite_2.3.1
##  [81] utf8_1.2.3                    tidyr_1.3.0
##  [83] generics_0.1.3                data.table_1.14.8
##  [85] rtracklayer_1.60.0            prettyunits_1.1.1
##  [87] httr_1.4.6                    htmlwidgets_1.6.2
##  [89] S4Arrays_1.0.4                uwot_0.1.16
##  [91] pkgconfig_2.0.3               gtable_0.3.3
##  [93] blob_1.2.4                    lmtest_0.9-40
##  [95] XVector_0.40.0                htmltools_0.5.5
##  [97] dotCall64_1.0-2               ProtGenerics_1.32.0
##  [99] scales_1.2.1                  png_0.1-8
## [101] knitr_1.43                    rstudioapi_0.14
## [103] rjson_0.2.21                  reshape2_1.4.4
## [105] curl_5.0.1                    nlme_3.1-163
## [107] zoo_1.8-12                    cachem_1.0.8
## [109] BiocVersion_3.17.1            KernSmooth_2.23-22
## [111] parallel_4.3.2                miniUI_0.1.1.1
## [113] AnnotationDbi_1.62.1          restfulr_0.0.15
## [115] pillar_1.9.0                  grid_4.3.2
## [117] vctrs_0.6.3                   RANN_2.6.1
## [119] promises_1.2.0.1              BiocSingular_1.16.0
## [121] dbplyr_2.3.2                  beachmat_2.16.0
## [123] xtable_1.8-4                  cluster_2.1.5
## [125] evaluate_0.21                 GenomicFeatures_1.52.0
## [127] Rsamtools_2.16.0              cli_3.6.1
## [129] compiler_4.3.2                rlang_1.1.1
## [131] crayon_1.5.2                  future.apply_1.11.0
## [133] labeling_0.4.2                plyr_1.8.8
## [135] stringi_1.7.12                viridisLite_0.4.2
## [137] deldir_1.0-9                  BiocParallel_1.34.2
## [139] munsell_0.5.0                 Biostrings_2.68.1
## [141] lazyeval_0.2.2                spatstat.geom_3.2-4
## [143] Matrix_1.6-1                  ExperimentHub_2.8.0
## [145] hms_1.1.3                     sparseMatrixStats_1.12.0
## [147] bit64_4.0.5                   future_1.33.0
```

```
## [149] KEGGREST_1.40.0              shiny_1.7.4.1
## [151] highr_0.10                   interactiveDisplayBase_1.38.0
## [153] AnnotationHub_3.8.0          ROCR_1.0-11
## [155] igraph_1.5.0.1               memoise_2.0.1
## [157] bit_4.0.5
```